

Lists and Loops – Practice Worksheet

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Sample Problem: Countdown

Write a program that counts down from 10 all the way to the number 1 and then prints blast off at the end

Example Output:

10...
9...
8...
7...
6...
5...
4...
3...
2...
1...
Blast off!!!

Solution:

```
count = 10
while count > 0:
    print(str(count) + "...")
    count -= 1
print("Blast off!!!")
```

Practise Problems

Problem 1: Character Slicer

Write a program that will take in a word from the user and then loop through each character in order to print each individual letter out until it has completed

Example Output:

Enter a word: Cheese

The word's characters separated individually are:

C

h

e

e

s

e

Problem 2: Sum of Numbers

Write a program that will find the sum of all the numbers between 1 and up to some user given value.

Hint: You may use for loops or while loops to accomplish this

Example Output:

Enter a number: 20

The sum of all numbers from 1 to 20 is: 210

Problem 3: Multiplication Table

Write a program that prints the multiplication table of a number n entered by the user (it should work up to 10).

Example Input:

Enter a number: 3

Example Output:

3 x 1 = 3

3 x 2 = 6

3 x 3 = 9

3 x 4 = 12

3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

Problem 4: Number Guessing Game

Using a while loop create a number guessing game, your task is to keep asking the user to guess some secret number between 1 and 10 until they are correct, to answer this question you are required to set the number that the user will guess before hand

Example Output:

```
Guess the number: 5
Try again!
Guess the number: 7
Correct!
```

Problem 5: The final challenge

It's time to put everything you've learned together!

Your challenge is to create a simple text-based game that uses loops and conditionals.

You decide what the game looks like — maybe you're a warrior in a dungeon, a space explorer, or a treasure hunter - maybe its just a day in you life that you want to gamify!, whatever the case may be, your but your program must use as many of the loops and concepts you have learned thus far.

Some Helpful Tips

- Use a while loop to keep the game running until a “game over” condition is met.
- Use if / else statements for decision-making (for example, fight or run).
- Use break to end the game when a certain condition happens (like losing all health or reaching a goal).
- Use continue if you want to skip parts of the loop (like dodging a trap or missing a turn).

Example Output (From my Game)

Welcome, hero! You enter the dark dungeon...

You have 3 health points.

Do you go left or right? left

You fight a goblin! You lose 1 health. Health: 2

Do you keep going? yes

You found treasure! You win!